



Salty survivability

Research from a team of scientists states that bacteria might be able to survive under freezing and salty environments. <https://digit.in/Aug18-63-2>



Kepler's death

The decade old planet hunter, Kepler Space Telescope, has discovered about 3,750 exoplanets and is now running low on fuel. <https://digit.in/Aug18-63-2>

only crude oil in terms of imported value, investment in gallium nitride fabrication is essential. Are we doing it as fast as possible? That's a slightly different story.

THE BUILD UP

The first ray of hope for this project was in 2009, when the Defence Research and Development Organisation (DRDO) gave Srinivasan a grant of 10 crore to set up a gallium nitride reactor at IISc. By then IISc already had some state of the art nanotechnology facilities, thanks to the Mission on Nanoscience and Technology, often called the Nano Mission, which was started in 2007. A part of this facility is a 1400 sqft room where the level of dust particles is a few millionth of that outside the facility. Not only does this give the researchers the capability to develop new semiconductor devices, it also allows them to test their applications successfully.



Gallium Crystals

The most difficult part of this project came in the initial days. Silicon tends to repel gallium nitride the way water repels oil, yet it was essential to grow nitride films on silicon wafers. To overcome this problem, the team decided to deposit gallium nitride as a series of thin layers, together which would count for thickness up to a millionth of a metre – and achieved success. Now all that was needed was to apply this technology at a larger scale.

SCALING UP

By the time the team at IISc had developed the technology to manufac-

ture gallium nitride laced chips in the country, GaN products were already trickling into the market. For the common people, one of the major applications that we see everyday is the stepping down of DC voltages powering smartphones and laptops. Another use-case seeing a rise in popularity is that of charging electric vehicles.

Manufacturing gallium nitride devices isn't expensive – it actually turns out to be cheaper, but requires considerably greater expertise. As the next step of their project, the team has submitted a proposal to the Ministry of Electronics and Information Technology (MeitY). As a part of this proposal, the team of researchers has outlined a plan that will cost the country ₹2500 crore over five years post approval for building the foundry. The foundry would ship gallium nitride wafers by the third year of operations. Radio frequency transistors, a strategic product, would be manufactured from the fourth year and power electronic transistors would be ready from the fifth year onwards.

That being said, the process is still not capable of being 100% indigenous. Some machinery, along with an important raw material, trimethyl gallium, will have to be imported – but the technology to take advantage of these will not.

PLAN B

While this proposal still awaits approval from MeitY, the team has already made alternate plans for potential funding. They intend to set up a company and seek investments of ₹300 crore to start off the factory. With the business that it will be able to satisfy at that point, it will attract more investment. With each level of complexity fulfilled at the factory, the value of the products will grow manifold. For instance, a normal silicon wafer ranges between ₹8,000 to ₹10,000, whereas adding the gallium nitride to the same wafer puts its value at ₹2-3

SILICON vs GALLIUM NITRIDE

While we've explained in brief why using gallium nitride to build electronics is advantageous, there are some specific reasons why that is the case:

- **Lower cost per function:** GaN devices use standard silicon substrates, hence there is no cost penalty. In fact, there is no true cost difference in terms of starting material per wafer. On the other hand, for a similar current carrying capability, a GaN device is smaller than a silicon device and hence achieves a lower cost per function.
- **Similar assembly costs, lower testing costs.**
- **Eliminates the need for a package:** Due to its structure, GaN on silicon eliminates the need for a package, hence reduces cost, wastage of board space while also doing away with the need for added electrical and thermal resistance without compromising on reliability.

lakh. If the development of power devices is successfully done, the same wafer can be sold for more than ₹50 lakhs. Hence, investment should be available in abundance. What we will lack in, though, is fabless chip design.

THE NEAR FUTURE

Gallium nitride fabrication units are running around the world now, with expertise from multiple universities and the industry working hand in hand with the government to create successful projects. India is too far behind on the silicon front to reasonably catch up any more. In case of gallium nitride, the race may not be that difficult and all it would take is a few years with the right investment planned. For all we know, when India goes full electric in terms of automobiles, all the converters and other components required might be manufactured entirely on our own soil! 